MILITARY SPECIFICATION

COMPUTER, BALLISTICS: M13A3

1. SCOPE

1.1 This specification covers one type of electro-mechanical analog ballistics computer. It is a component of the primary fire control system for the 105 Millimeter Gun, Tank M48A2C.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Ordnance Material, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Frankford Arsenal

F7650085	Fixture, Vibrator
F7659607	Fixture
F7659608	Adapter
F10527323	Computer, Ballistics: M13A3

PACKAGING DATA SHEET

MIL-P-14232/P8619966	Packaging of Computer, Ballistics

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions

FSC 1220

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should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample.- Requirements for the submission of first article samples by the contractor for testing shall be as specified in the contract.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction.- The Ballistics Computer M13A3 shall be manufactured in accordance with Drawing F10527323 and drawings pertaining thereto.

3.4 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship

3.5 Performance.

3.5.1 Output calibration.

3.5.1.1 Zero reference.- Input shaft rotation shall be measured from a precise 1200 meter zero reference check point previously determined by the manufacturer and indicated by an index line inscribed on the input shaft in coincidence with an index on the input shaft seal cover plate.

3.5.1.2 1200 meter check point.- With the computer having a constant load of between 75 and 90 ounce-inches applied to the output shaft for ballistics drive, the ammunition selector shall be set on the APDS-T, M392 Cam. With the range correction knob set to zero and with the input shaft set at the 1200 meter check point, actuate the superelevation hand crank to match the outer index with the inner index within plus or minus half a line width, by approaching this range first from one direction then from the opposite direction without over travel. Note the position of the output shaft for ballistics drive and split the backlash error at this output shaft. This is the check point for computer output. The position of the superelevation counter shall be established from this check point and shall conform to the requirements of Table I.

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TABLE I

Position of Superelevation Counter

Output Shaft Position	117.36° ccw	1200 meter check point 0°	1R-8.64° cw	1R-296.64° cw
Tolerance	±8°	±3°	±8°	±8°
Superelevation Counter Mils	1.25	2.88	5.00	12.00

3.5.1.3 Outer index and inner index.- Using the APDS-T, M392 Cam only, position the input shaft to correspond with each of the ranges as indicated in Table II and rotate the hand crank until the outer index is in coincidence with the inner index, at which point the rotation of this output shaft shall correspond to the values in Tables II and III. This procedure shall be repeated to check each cam at 1200 meters only. All checks shall be made by approaching the index from both clockwise and counterclockwise direction.

3.5.1.4 Backlash.- Backlash of the output shaft for the superelevator shall not exceed a total of one degree at any point of rotation of the shaft.

3.5.1.5 Outer index and range scale.- Using the APDS-T, M392 Cam only, rotate the hand crank until the outer index is in coincidence with the selected range on the computer range scale for each of the ranges as indicated in Table II, at which point the rotation of the output shaft for ballistics drive shall correspond to the values in Tables II and III. All checks shall be made by approaching the index from both clockwise and counterclockwise direction.

3.5.1.6 Servo follow-up sensitivity.- With an output shaft load of between 75 and 90 ounce-inches applied to the output shaft for the ballistics drive; the ammunition selector shall be set on the APDS-T, M392 Cam; the power on at 18 ± 0.5 volts, and the automatic follow-up cycle completed at ranges between 3800 and 4400 meters, rotate the input shaft slowly so that it is possible to count the number of times the contact closes in order to advance the output shaft to the ballistics drive by approximately 72 degrees. Calculate the average change in superelevation per one make and break of the contacts. Repeat in the opposite direction. The average allowable superelevation increment for each range shall not exceed 10 degrees 45 minutes and the maximum superelevation increment shall not exceed 21 degrees 30 minutes as read at the output shaft to the ballistics drive.

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3.5.2 Power operation.- With the computer having a constant load of between 75 and 90 ounce-inches applied to the output shaft for the ballistics drive, the rotation of this output shaft shall correspond to the values indicated in Tables II and III for the ammunition and ranges specified therein. This requirement shall be met by allowing the outer index to approach the inner index from both increasing and decreasing range with power on continuously at 28 ± 0.5 volts. The computer shall also meet the accuracy requirement of Tables II and III for the following ammunition and at ranges indicated when checked at 18 ± 0.5 volts.

Ammunition and Range for 18 Volt Check

Ammunition	
APDS-T, M392	500, 1000, 2000 and 3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456, (T384E4)	500, 1000, 2000 and 3600 meters

All checks shall be made by approaching index from both clockwise and counterclockwise direction.

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TABLE II

Computer Accuracy as Determined by Position of Rotation of the Output Shaft Based on 1200 Meter Reference Range for APDS-T, M392 Cam (Drawing No. 8619776).

RANGE METERS	500	1000	1200	1500	2000	3000	3600	4400
INPUT SHAFT POSITION (±1°)	8R 330.48° CW	1R 308.52° CW	0° CHECK POINT	2R 98.25° CCW	5R 73.08° CCW	9R 119.88° CCW	11R 68.40° CCW	13R 84.24° CCW
APDS-T	125.28° CCW	36.72° CCW	0°	56.88° CW	152.64° CW	357.12° CW	1R 131.76° CW	1R 339.12° CW
M392	*1.14	*2.37	*2.88	*3.67	*5.00	*7.84	*9.71	*12.59
HEP-T	273.60° CW	2R 21.60° CW	2R 239.04° CW	3R 244.80° CW	5R 254.16° CW	11R 101.52° CW	15R 284.40° CW	
M393	*6.68	*13.18	*16.20	*21.28	*31.41	*59.29	*81.83	
HEAT-T	65.52° CCW	105.12° CW	182.88° CW	312.48° CW	1R 208.08° CW	3R 240.48° CW	5R 228.96° CW	9R 289.44° CW
M456 (T384E4)	*1.97	*4.34	*5.42	*7.22	*10.77	*21.22	*31.06	*51.90

(*)Corresponding nominal superelevation mil value. (See 6.3)

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3.5.2.1 Output shaft tolerances.- When the output shaft rotation of the computer, as read in degrees, is equivalent to the superelevation value in mils as shown in Table III, column A, the output shaft tolerance shall be within the corresponding value as shown in column B.

TABLE III

Output Shaft Tolerances

A Output Shaft Rotation	B Degrees	C Mils*
0 up to but less than 6	± 7.20	±0.10
6 up to but less than 11	±10.80	±0.15
11 up to but less than 26	±14.40	±0.20
26 up to but less than 31	±18.00	±0.25
31 up to but less than 36	±25.20	±0.35
36 up to but less than 41	±28.80	±0.40
41 up to but less than 61	±36.00	±0.50
61 up to but less than 86	±43.20	±0.60
86 up to but less than 98.5	±50.40	±0.70

(*) The data contained in column C is informational only. (See 6.3)

3.5.3 Stability.- Using 28 ± 0.5 volts, with a load of between 14 and 18 ounce-inches applied to the output shaft to ballistics drive, ballistics correction set to introduce minus 15 percent within 1 percent, the rotation of the input shaft between minimum range and 1200 meters shall not cause the computer servo to display any self-sustaining or oscillatory operation. This check is required for all cams. A maximum of eight changes in direction of the servo motor, as a settling out action, shall be considered as satisfactory operation. These eight changes may be counted as clicks.

3.5.4 Component operation.

3.5.4.1 Operational values shall not exceed those given in Table IV at the temperatures indicated.

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TABLE IV

Operational Values

Control	Operation	Room Temp. to 150°F.	Minus 40°F
Input Shaft	Rotate	16 oz.-in.	45 oz.-in.
Range	Pull Out	20 lbs.	24 lbs.
Correction Knob	Rotate	34 oz.-in.	45 oz.-in.
Ammunition	Rotate	60 lb.-in.	80 lb.-in.
Selector	Translate	10 lbs.	20 lbs.
Handcrank	Hold Engaged	12 lbs.	20 lbs.
	Rotate Engaged	6 lb.-in.	20 lb.-in.
Reset Button	See Par. 3.6.3	15 lbs.	40 lbs.
Iris Dimmer	Rotate	Mechanically Operable	
Interrupter Switches	See Par. 3.6.3	Mechanically and Electrically Operable	

3.5.4.2 Time of response.- At an ambient temperature (60° to 90° Fahrenheit); the output shaft load as specified in 3.5.2 applied; the inner and outer indexes positioned to the values in Table V; and the ammunition selector set on the APDS-T, M392 Cam, the outer index shall stabilize on the inner index within the elapsed time specified after power has been applied.

TABLE V

Time of Response

Outer Index Position (Meters)	Inner Index Position (Meters)	Voltage (Plus or Minus 0.5V)	Total Elapsed Time (Seconds)
4400	500	28	4
500	4400	28	4
500	4400	18	8
4400	500	18	8

3.5.4.3 Circuit breaker.

3.5.4.3.1 Breaker opens.- With the motor at an ambient temperature between 60° to 90°F and with 28 ± 0.5 volts applied; the two indexes separated by at least 200 meters, and the output shaft for the ballistics drive locked (immovable), the circuit breaker shall open in 90 seconds or less.

3.5.4.3.2 Breaker closed.- At an ambient temperature between 60° to 90°F; the load as specified in 3.5.2 applied to the output shaft for the ballistics drive; the computer operating at 28 ± 0.5 volts, using the APDS-T M392 Cam; the input shaft shall be oscillated for a period of forty-five seconds over a spread of 1000 ± 100 meters between ranges of 1200 to 2800 meters at a rate of 4 to 4.5 revolutions per second, during

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which time the circuit breaker shall not open.

3.5.4.4 Hand crank and input shaft travel.

3.5.4.4.1 The rotation of the superelevation hand crank within the limits of its stops shall cover the excursion from minus 1.5 mils ± 0.1 mils to plus 98.5 ± 0.5 mils as indicated on the superelevation counter when the ballistics correction knob is set at zero.

3.5.4.4.2 The range input shaft shall rotate through an excursion of 24.0 revolutions plus 0.1 revolution, and shall cover the range from 480 to 4575 meters within this excursion, as measured from the 1200 meter check point of the input shaft. (See 6.4)

3.5.5 Superelevation cams.- The cams shall be placed in the ballistics computer in the following sequence starting with the APDS-T M392, followed by: the HEP-T, M393; the HEAT-T, M456 (T384E4); and three spacers. The computer shall be shipped with the same cams with which it was qualified.

3.6 Electrical check.

3.6.1 Insulation breakdown.- Insulation breakdown shall be in accordance with MIL-F-13926 except that breakdown voltage of 500 volts root mean square (rms) shall be substituted for 750 volts rms.

3.6.2 Insulation resistance.- Insulation resistance shall be in accordance with MIL-F-13926 except that the resistance value shall not be less than 2 megohms and a direct current (dc) potential of 500 volts shall be used in place of 800 volts. All lamps shall be removed from sockets prior to tests.

3.6.3 Continuity.- The electrical circuits shall be checked for continuity and shall conform to the wiring diagram. When the hand crank is engaged, the interrupter switch shall open the motor circuit and energize the pilot light. Pressing of the reset button shall cause the motor circuit to be closed and the pilot light to be extinguished. Actuation of the ammo selector handle shall cause the ammunition selector interrupter switch to open the motor circuit. Resetting of the handle shall cause the switch to complete the circuit.

3.6.4 Illumination.- The internal lighting device shall provide ade uate illumination so that the inner and outer indexes of the range dial and the ammunition index shall be clearly defined and readable when observed in a darkened enclosure.

3.7 Environmental.

3.7.1 Storage temperatures.- The computer shall show no evidence of physical failure and shall meet the re uirements of 3.7.2.1 at an ambient temperature of 60° to 90°F after having been thermally stabilized at temperatures of minus 80° and plus 160°F.

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3.7.2 Extreme temperature operability.- The computer shall meet the following requirements while thermally stabilized at temperatures of minus 40° and plus 150° F:

(a) Applicable requirements of 3.5.4.1 at temperatures indicated in Table IV.

(b) At minus 40°F, the outer index shall follow the inner index within one-half a graduation width over the full excursion of the range scale in a total elapsed time of 8 seconds for each direction when checked with 18 ± 0.5 volts.

(c) At plus 150°F using 28 ± 0.5 volts, the ammunition selector set on the APDS-T, M392 Cam, with a load of between 14 and 18 ounce-inches applied to the output shaft for the ballistics drive and with the indexes separated by 3600 meters, the outer index shall stabilize on the inner index and shall meet the applicable requirements as specified in 3.5.3 when the switch is thrown to the ON" position. Immediately upon completion of the environmental requirements as specified in 3.7, the interior of the computer shall be dried by clean, warm air between 100° to 140°F.

3.7.2.1 Post environmental check.- The computer shall meet the following requirements while thermally stabilized between 60° to 90°F:

(a) The requirements of Table I and 3.5.3.

(b) When checked under power operation at 28 ± 0.5 volts, with 75 to 90 ounce-inches load applied, the output shaft for the ballistics drive shall indicate the rotational values of Table II within the tolerances of Table III for the following ammunition and ranges:

Ammunition	Ranges
APDS-T, M392	500, 1000, 2000 and 3600 meters
HEP-T, M393	500, 1000, 2000 and 3600 meters
HEAT-T, M456 (T384E4)	500, 1000, 2000 and 3600 meters

3.7.3 Vibration.- With the ammunition selector set on the APDS-T, M392 Cam, the outer index set at 3000 meters, the computer (less any shock mounts and brackets, Universal Assembly B8200862 and Output Shaft B8200801) shall be vibrated at an amplitude of 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes ± 15 seconds.

3.7.3.1 Post vibration.- Subsequent to vibration, the computer shall not be damaged or show evidence of broken, loose or missing parts and it shall meet all the requirements of this specification.

3.7.4 Sealing.- The computer shall meet the requirements of the rain test of MIL-F-13926.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

4.2.1 Preproduction test.- When specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4, and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926, and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP). Examinations and tests related to Section 3

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herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables VI, VII, and VIII herein. Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE VI - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE			
MAJOR: AQL 1.0% defective			
101.	Physical failure, post vibration	3.7.3.1	4.6.4.1
102.	Output shaft	3.5.1.2	4.6.5.1
103.	Outer index and inner index	3.5.1.3	4.6.5.2
104.	Backlash	3.5.1.4	4.6.5.3
105.	Outer index and range scale	3.5.1.5	4.6.5.4
106.	Servo follow-up sensitivity	3.5.1.6	4.6.5.5
107.	Power operation and output shaft tolerance	3.5.2	4.6.5.6
108.	Stability	3.5.3	4.6.5.7
109.	Time of response	3.5.4.2	4.6.5.8
110.	Circuit breaker closed	3.5.4.3.2	4.6.5.9.1
111.	Circuit breaker opens	3.5.4.3.1	4.6.5.9.2
112.	Hand crank travel	3.5.4.4.1	4.6.5.10.1
113.	Input shaft travel	3.5.4.4.2	4.6.5.10.2
114.	Superelevation cams	3.5.5	4.6.5.11
115.	Operating values	3.5.4.1	4.6.6
116.	Insulation breakdown	3.6.1	4.6.7.1
117.	Insulation resistance	3.6.2	4.6.7.2
118.	Continuity	3.6.3	4.6.7.3
119.	Illumination	3.6.4	4.6.8
MINOR: NONE			

NOTE: The above examinations shall be conducted at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

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4.4.1 General.- One computer shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table VII.

TABLE VII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301. Construction	3.3	Applicable drawing-visual
302. General specification	3.4	MIL-F-13926-visual and tactile

4.4.2 Environmental.- Three computers shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table VI and shall then meet the requirements and tests in Table VIII.

TABLE VIII

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
303. Stability (-40°F. and +150°F.)	3.7.2	4.6.1.2
304. Outer index and inner index (-40°F.)	3.7.2(b)	4.6.1.3
305. Outer index and inner index (+150°F.)	3.7.2(c)	4.6.1.4
306. Operating torques (-40°F. and +150°F.)	3.7.2(a)	4.6.1.5
307. Physical failure, post environmental	3.7.1	4.6.1.6
308. Output shaft, post environmental	3.7.2.1 3.5.3	4.6.2.1
309. Stability, post environmental	3.7.2.1 3.5.3	4.6.2.2
310. Power operation, post environmental	3.7.2.1	4.6.2.3
311. Sealing (water resistance)	3.7.4	4.6.3

NOTE: The tests for characteristics 303, 304, 305 and 306 shall be conducted at the temperatures indicated. The tests for characteristics 307, 308, 309, 310 and 311 shall be conducted at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit) subsequent to the temperature cycling test.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents (IEL), when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the p·oduct tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Environmental test.- Samples for environmental testing shall be selected as specified. The contractor's Government approved refrigeration, heating and rain test chambers shall be used to perform the environmental test.

4.6.1.1 The operational temperature test may be conducted concurrent with the storage temperature test. Caution must be exercised so as not to expose the computer to thermal shock during the test. Temperature changes shall be gradual.

4.6.1.2 Stability.- The computer shall be subjected to this test when thermal equilibrium of the computer is established at minus 40 degrees and plus 150 degrees Fahrenheit, respectively. With the ammunition selector set on the cam as specified in 3.5.1.2, this test shall be conducted as specified in 4.6.5.7. The computer shall meet the requirements as specified in 3.5.3 and 3.7.2.

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4.6.1.3 Outer index and inner index.- When thermal equilibrium of the computer is reached at minus 40 degrees Fahrenheit, the computer outer index shall meet the specified requirements when the inner index is translated over the full excursion of the range scale within the time limit specified. This translation of the inner index shall be accomplished by a continuous rotation of the computer input shaft from the clockwise stop to the counterclockwise stop. Repeat test by rotating the input shaft in the opposite direction. This test shall be conducted with the specified voltage applied to the computer and the circuit breaker switch in the "ON" position.

4.6.1.4 Outer index and inner index.- When thermal equilibrium of the computer is reached at plus 150 degrees Fahrenheit, set the ammunition selector on the cam specified, the range correction knob to zero and apply the specified torque load to the computer output shaft. Position the computer inner index at 500 meters and the outer index at 4400 meters on the range scale. When the circuit breaker switch is thrown to the "ON" position, the outer index shall "settle out" (cease oscillatory operation) coincident to the inner index within eight (8) changes (clicks) in the direction of the computer servo. The computer shall again meet this requirement when the test is repeated with the outer index positioned at 500 meters and the inner index positioned at 4400 meters on the range scale.

4.6.1.5 Operating torques and force.- When thermal equilibrium of the computer is reached at minus 40 degrees and plus 150 degrees Fahrenheit, respectively, the controls listed in Table IV shall not exceed the applicable torque and force values specified therein for the operation of those controls. This test shall be conducted as specified in 4.6.6.

4.6.1.6 Physical failure, environmental.- Each computer subjected to the storage temperatures shall be examined for physical failure of castings, mechanical parts, electrical parts and glass breakage. This examination shall be conducted at standard ambient temperature after exposure to each temperature extreme.

4.6.1.7 Drying.- Immediately following the environmental test, the computers subjected to that test shall be air dried as specified.

4.6.2 Post environmental check.- Subsequent to the environmental check, the computer subjected to that test shall be set up as specified in 4.6.5.

4.6.2.1 Output shaft, post environmental.- This test shall be conducted as specified in 4.6.5 and 4.6.5.1 and shall meet the requirements of Table I.

4.6.2.2 Stability, post environmental.- This test shall be conducted as specified in 4.6.5 and 4.6.5.7 and shall meet the specified requirements.

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4.6.2.3 Power operation, post environmental.- This test shall be conducted as specified in 4.6.5, 4.6.5.6 and 4.6.5.6.1 for the cams and ranges specified. The results of this test shall meet the applicable requirements of Tables II and III.

4.6.3 Sealing (water resistance).- This test shall be conducted in accordance with Procedure I of MIL-F-13926. The computer shall show no evidence of water penetration as a result of this test. The Receptacle 8200970 shall be covered with waterproof tape during this test.

4.6.3.1 The computers that are opened for environmental drying and examination for water penetration shall be resealed to meet the requirements of the drawings prior to submission for final acceptance.

4.6.4 Vibration.- Each computer in the sample lot shall be vibrated as specified in 3.7.3. This operation shall be performed on Fixture F7650085 using Adapter F7659608.

4.6.4.1 Physical failure, post vibration.- As a result of vibration, the computer shall show no evidence of physical failure of mechanical parts or glass breakage.

4.6.5 The following tests shall be conducted with the computer positioned on Test Fixture F7659607 and in accordance with the operating instructions specified on Drawing F7659607. The computer input shaft and the fixture input mechanism shall be oriented to the 1200 meter check point as outlined in the set-up instructions on Drawing F7659607.

4.6.5.1 1200 meter check point.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the mil value specified in Table I is indicated on the computer superelevation mil counter, the rotational position of the output shaft for the ballistics drive shall not exceed the tolerance specified in Table I.

4.6.5.2 Outer index and inner index.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the rotational position of the computer input shaft corresponds to the value for each of the ranges indicated in Table II, and the computer outer and inner indexes are coincident, the rotational position of the ballistics drive output shaft shall not exceed the tolerance specified for the applicable values in Tables II and III. Each cam shall meet the requirements of Tables II and III for the specified range.

4.6.5.3 Backlash.- Backlash of the computer superelevation output shaft shall not exceed the tolerance when measured at four (4) points of shaft rotation. The four (4) points of measure shall be approximately 90 degrees apart.

4.6.5.4 Outer index and range scale.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the computer outer index is coincident to the selector range mark on the computer range scale, the rotational position of the output shaft for the

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ballistics drive shall not exceed the tolerance specified for the applicable values in Tables II and III.

4.6.5.5 Servo follow-up sensitivity.- Set the ammunition selector on the cam specified, the range correction knob on zero and the inner and outer index to a range between 3800 meters and 4400 meters. With the computer in power operation, the average and maximum recorded increment change (closing of computer servo contacts) resulting from a total rotation of the output shaft 72 degrees clockwise and 72 degrees counter-clockwise shall not exceed the tolerance specified. Superelevation increment changes shall be measured at the computer output shaft for the ballistics drive.

4.6.5.6 Power operation.- This test shall be conducted as specified in Drawing F7659607 with the range correction knob set to zero.

4.6.5.6.1 With the input voltage set at 28 volts, plus or minus 0.5 volts, and the circuit breaker switch in the "ON" position, the accuracy of each cam shall be tested at the ranges listed in Table II for computer output shaft rotation. When the input shaft is rotated to a value indicated in Table II for a given range, the rotational position of the output shaft shall correspond to the value for that range in Table II within the tolerance specified in Table III.

4.6.5.6.2 With the input voltage set at 18 volts, plus or minus 0.5 volts, the computer shall meet the accuracy requirements of Tables II and III for the cams and ranges.

4.6.5.7 Stability.- Prepare the computer for this test in accordance with Drawing F7659607, set the range correction knob to minus 15 percent within 1 percent and adjust the input voltage to 28 volts, plus or minus 0.5 volts. Rotate the computer inner index 1500 meters; throw the circuit breaker switch to "ON" position to energize the servo motor. The outer index shall "settle out" (cease oscillatory operations) coincident to the inner index within 8 changes (clicks) in direction of the computer servo. The inner index shall then be rotated to 4 different ranges between 1500 meters and minimum range, and the outer index again shall "settle out" coincident to the inner index within 8 changes in direction of the servo. Set the range correction knob to plus 15 percent within 1 percent and repeat the aforementioned test. All cams installed in or accompanying the computer shall be subjected to this test.

4.6.5.8 Time of response.- Set the ammunition selector on the cam specified and the range correction knob on zero. When the circuit breaker switch is thrown to the "ON" position and the servo motor is energized, the computer outer index shall stabilize on the inner index within the time cycle specified in Table V.

4.6.5.9 Circuit breaker.

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4.6.5.9.1 Circuit breaker, closed.- Set the ammunition selector on the cam specified, the range correction knob on zero and the inner and outer index coincident on 2000 meters of range. The computer input shaft shall be oscillated as specified with the circuit breaker switch in the "ON" position and the servo motor energized. The circuit breaker shall not opern (switch move to "OFF" position) as a result of this oscillation.

4.6.5.9.2 Circuit breaker, opens.- Set the ammunition selector and range correction knob as specified in 4.6.5.9.1. The inner and outer index shall be separated by the specified distance. When the circuit breaker switch is thrown to the "ON" position and the servo motor is energized, the circuit breaker shall open (switch move to "OFF" position) within the specified time cycle.

4.6.5.10 Hand crank and input shaft travel.

4.6.5.10.1 The test for hand crank travel shall be conducted in accordance with Drawing F7659607 and shall meet the specified requirements.

4.6.5.10.2 The test for input shaft travel shall be conducted in accordance with Drawing F7659607 and shall meet the specified requirements.

4.6.5.11 Superelevation cams.- The inspector sequence and arrangement of cams in the computer shall be as specified. The process of replacing cams in the computer shall be as specified in Drawing F7659607.

4.6.6 Operating torques and force.- The controls listed in Table IV shall not exceed the applicable torque and force values specified therein for the operation of those controls while at an ambient temperature of plus 60 degrees to 90 degrees Fahrenheit. The equipment to be utilized for this test is listed on Drawing D8570433. The reset button shall be inspected by use of standard measuring equipment. This standard measuring equipment shall be capable of measuring the values within one-quarter (1/4) pound.

4.6.7 Electrical check.- The following tests shall be conducted at any time subsequent to vibration. Before testing, remove lamp bulbs and disconnect capacitor from computer circuit.

4.6.7.1 Insulation breakdown.- This test shall be conducted in accordance with MIL-F-13926 and shall meet the specified requirements Procedure I shall apply.

4.6.7.2 Insulation resistance.- This test shall be conducted in accordance with MIL-F-13926 and shall meet the specified requirements.

4.6.7.3 Continuity.- A continuity test of the computer circuits shall be made using standard measuring equipment to determine conformance with the wiring diagram. Actuation of the hand crank, reset button and ammunition selector handle shall cause the computer to function.

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4.6.8 Illumination.- The illuminated inner and outer indexes of the range dial and ammunition index of the computer shall meet the requirements when observed in a darkened enclosure with the unaided eye.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The computer shall be packaged, packed and marked in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8619966. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- Ballistics Computer M13A3 is intended to receive log range data from a range finder calibrated for range measurement in meters. The computer is intended to employ log range data to manually or electrically compute and relay to the (105MM Gun, Tank M48A2C) sighting system the amount of gun superelevation required for a specific target range and ammunition type.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Order of document precedence.

6.3 Superelevation values in mils.- The mil values in Table II represent the corresponding nominal superelevation values in mils for the calculated rotational values in the table. The actual superelevation value in mils that will correspond to computer output rotation will be obtained by modifying the nominal value by the relationship that each 7.20 degrees of rotation of the output shaft above or below the nominal value is equal to 0.10 mil.

6.4 Input shaft excursion.

From 1200 meter zero check point

9R	119.97°	CW	=	480 meters
13R	84.24°	CCW	=	4400 meters
13R	227.40°	CCW	=	4575 meters

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1220-A241Y

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